

Monday Warm Up: Unit 1, Capacitors, Current and DC circuits

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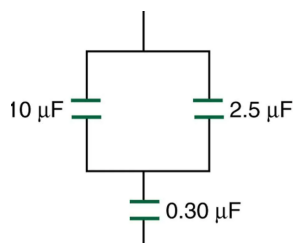


Figure 1: Two capacitors *in parallel* (top), that are *in series* with a third (bottom).

1 Memory Bank

- $\Delta U = q\Delta V$... Relationship between change in potential energy, charge, and voltage.
- $V(x) = Ex + V_0$... Linear relationship between voltage, electric field, and distance between two charge plates to form a capacitor.
- $Q = C\Delta V$... Relationship between stored charge, capacitance (C), and voltage. Capacitance has the unit of 1 Farad, or $1\ \text{C}\ \text{V}^{-1}$.
- $I = \Delta Q / \Delta t$... The relationship between current (I), charge, and time. The unit of current is 1 Amp (A), or $1\ \text{C}\ \text{s}^{-1}$.
- $\Delta U = \frac{1}{2}C\Delta V^2$... The energy stored in a capacitor.
- $C = \epsilon_0 A / d$... The capacitance of a parallel-plate capacitor with area A and plate separation d .

2 Capacitors and Current

1. (a) Suppose we allow 5 mA of current to flow onto a capacitor for 5 seconds. How much charge is stored in the capacitor? (b) After 5 seconds, the capacitor voltage is measured to be 5.0 V. What is the capacitance?

2. What capacitance is needed to store $3.00\ \mu\text{C}$ of charge at a voltage of 120 V?

3. (a) Find the capacitance of a parallel plate capacitor having plates of area $4.00\ \text{cm}^2$ that are separated by 0.100 mm. (b) Filling the space between the plates with Teflon increases the value of ϵ_0 by a factor of 2.1. If the capacitor in part (a) was filled with Teflon, what would the new capacitance be?

4. Consider Fig. 1. Suppose a battery is connected to the top and bottom of the circuit. The first two capacitors are *in parallel*, because they would split the current, but share the voltage. Their total capacitance is

$$C_{\text{pair}} = C_1 + C_2 \quad (1)$$

- (a) What is the total capacitance of the top two capacitors? (b) The top pair of capacitors are *in series* with the lower one, because they share the current but split the voltage. The total capacitance is

$$C_{\text{tot}}^{-1} = C_{\text{pair}}^{-1} + C_3^{-1} \quad (2)$$

- (b) What is C_{tot} ?

3 Bonus Problem

1. A battery-operated car utilizes a 12.0 V system. Find the charge the batteries must be able to move in order to accelerate the 750 kg car from rest to $25.0\ \text{m}\ \text{s}^{-1}$, make it climb a 200 m high hill and then cause it to travel at a constant $25.0\ \text{m}\ \text{s}^{-1}$ by exerting a 500 N force for an hour. *Hint: the power to move an object at constant velocity $\vec{v}$ when requiring a force $\vec{F}$ is $P = \vec{F} \cdot \vec{v}$. Can you prove this?*